

Shivani Bokka

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TECHNICAL SKILLS

Applied AI & Machine Learning: Supervised & Unsupervised Learning, Deep Learning, Transfer Learning, Reinforcement Learning, NLP, Computer Vision, Generative AI, Agentic AI, LLMs, RAG, Fine-tuning (LoRA, QLoRA, DPO), Quantile Forecasting, Model Compression (Quantization, Pruning, Knowledge Distillation)

Agentic AI & LLM Stack: LangGraph, ReAct Reasoning Loops, MCP (Model Context Protocol), Anthropic Claude, OpenAI GPT, Groq, Mistral, Ollama, Gemini, Prompt Engineering, LLM Evaluation (BERTScore, G-Eval, LLM-as-Judge), Hallucination Detection, ChromaDB, FAISS, sentence-transformers

Deep Learning Frameworks & Models: PyTorch, PyTorch Lightning, TensorFlow, Keras, HuggingFace Transformers, PEFT, pytorch-forecasting (TFT), RoBERTa, DeBERTa-v3, Llama, ViT, CLIP, LSTM, PatchTST, NLTK, BERTopic

MLOps & Deployment: MLflow, Docker, Kafka, Prefect 2, Apache Airflow, Great Expectations, ONNX, Prometheus, Grafana, Feast, Redis, Drift Detection (KS test, PSI, Evidently), AWS (EC2, S3, SageMaker), Google Cloud Platform (AI Platform, BigQuery)

ML Libraries & Statistical Modeling: scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, GraphSAGE, K-Means++, GMM, CausalML (Uplift Modeling), SMOTE, Bootstrap CI, Walk-Forward CV, Bayesian Optimization, Isotonic Calibration

Programming, Data & Visualization: Python, SQL, MS SQL Server, MySQL, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Tableau, Power BI, Jupyter Notebooks, Git, AWS (EC2, S3, SageMaker), Google Cloud Platform (AI Platform, BigQuery)

PROFESSIONAL EXPERIENCE

Teaching Assistant (Data Science & Machine Learning Program/SFSU)

Aug 2025 – Present | San Francisco, CA

- Lead 15 weekly sessions for 30+ Genentech employees, teaching ML fundamentals including bias-variance tradeoff, cross-validation, and regularization with Python, NumPy, and PyTorch
- Guide learners in building deep learning models using CNNs and transformers, implementing custom training loops and loss functions for biotechnology analytics projects
- Develop weekly coding examples demonstrating testable code practices using Pandas and scikit-learn; improved average evaluation scores by 15% through hands-on debugging sessions

RELEVANT PROJECTS

Supply Chain Intelligence with MCP & LLM | Agentic AI

MCP, ReAct, TFT, CatBoost, GraphSAGE, Gradio — *GitHub: SCM-using-MCPand-LLM*

- Built a production-grade agentic system with a 50-tool MCP server and a ReAct agent routing across 4 LLM providers (Claude, OpenAI, Groq, Mistral) for inventory management, supplier scoring, and cold chain monitoring
- Implemented dual forecasting (TFT + CatBoost), GraphSAGE fraud ring detection, sliding window memory compression, and an 18-chart real-time Gradio dashboard

Fine-Tuned Domain LLM with QLoRA & DPO | AI Engineering

HuggingFace, QLoRA, DPO, MLflow, Gradio — *GitHub: Fine-Tuned-Domain-LLM-QLoRA*

- Fine-tuned Llama 3.2 3B on 510 legal contracts via 3-stage ablation (LoRA → QLoRA 4-bit NF4 → DPO), reducing GPU memory by ~60% while preserving task accuracy across clause extraction and contract analysis
- Evaluated with BERTScore, G-Eval, NLI-based hallucination rate, and ECE calibration; surfaced failure modes via UMAP + HDBSCAN clustering and deployed a FastAPI serving endpoint with MLflow experiment tracking

Production Fraud Detection System | ML Engineering

XGBoost, GraphSAGE, Redis, ONNX — *GitHub: Fraud-Detection-System*

- Designed a 3-layer fraud detection stack (rules → ML → human review) achieving AUC-ROC 0.997, and 93% dollar capture rate on 1.85M transactions, mirroring Stripe/Uber/PayPal architecture
- Engineered Redis sliding-window velocity features (5-time horizons), GraphSAGE entity embeddings for fraud ring detection, FP-Growth auto rule mining, and exported XGBoost to ONNX for low-latency serving

Customer Segmentation & Churn Uplift Engine | Data Science

CausalML, XGBoost, Streamlit — *GitHub: Customer-Segmentation-Churn-Engine*

- Segmented 5,630 customers via K-Means++ with GMM soft assignments; applied CausalML uplift modeling (T-Learner + S-Learner) to isolate 710 Persuadables — customers who respond to retention intervention vs. those who churn regardless
- Trained per-segment XGBoost classifiers with isotonic calibration, generated LLM-powered (Claude) retention recommendations, and delivered a 4-page Streamlit dashboard with interactive UMAP visualization and MLflow tracking

EDUCATION

San Francisco State University, San Francisco, CA

Master of Science, Data Science and Artificial Intelligence

Aug 2024 - Aug 2026

GPA: 3.88

Deenbandhu Chhotu Ram University of Science and Technology, India

Bachelor of Technology, Electronics and Communication

May 2011 - May 2016